

Mirror Quantum Awesomeness

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Quantum processing units now routinely exceed 100 qubits, but scalable performance assessment remains challenging, particularly for guiding near-term algorithms. Mirror Randomized Benchmarking provides one way to estimate QPU-scale fidelity by using randomized mirror circuits with Pauli twirls and state preparations, yielding average-layer infidelity via the exponential decay of the effective polarization. However, a primary disadvantage of this approach is the extensive sampling overhead that arises as system size increases. To address this, a novel variant of MRB is introduced in which circuits generate a set of entangled pairs rather than returning a single bit string. This modification allows the extraction of the same metrics as standard MRB while enabling alternative analyses with constant overhead, providing deeper insights into the performance of near-term quantum algorithms. Experiments on IBM Quantum hardware with over 100 qubits are used to validate and test the method, as well as to benchmark the hardware.

I. INTRODUCTION

The advent of quantum processing units (QPUs) surpassing 100 qubits marks a pivotal era in computational science, in which quantum systems begin to tackle problems intractable for classical computers, such as simulating molecular interactions or optimizing complex processes [1, 2]. The inherent parallelism provided by superposition, combined with interference effects, allows quantum systems to converge efficiently on a solution. [3]. However, realizing this potential is hampered by inherent errors, crosstalk, coherent noise, drift, and decoherence, which increase with system size and circuit depth, often rendering large-scale computations unreliable [4]. Traditional metrics, such as gate fidelities, offer localized insights but fail to predict holistic performance, underscoring the need for comprehensive benchmarking that captures how errors manifest in realistic workloads [5, 6].

Benchmarking serves as a bridge between hardware capabilities and algorithmic utility, providing actionable data for optimization in the noisy intermediate-scale quantum (NISQ) regime [7, 8]. Early protocols focused on customized random circuits to probe entanglement dynamics, but recent innovations emphasize scalability and interpretability. Among these, Mirror Randomized Benchmarking (MRB) [9] stands out for its ability to estimate average layer infidelities at QPU scales. MRB employs randomized mirror circuit sequences of Clifford layers, combined with Pauli twirls and local Clifford preparations, to measure effective polarization decay and approximate the infidelity of Pauli-dressed n -qubit layers. Simulations on up to 225 qubits with physically realistic error rates in the range 0.1%–1%, together with experiments on up to 16 physical qubits, demonstrate that MRB can detect crosstalk through infidelity growth with qubit count while maintaining robustness under Markovian errors [9]. However, its reliance on global ensemble

averaging leads to high sampling overhead for large systems, limiting fine-grained diagnostics.

Complementing the quantitative focus of MRB’s is the qualitative accessibility of Quantum Awesomeness (QA) [10, 11], a benchmarking protocol based on random circuits that generate sets of entangled qubit pairs. QA is designed not only as a quantitative benchmark but also as a visualization-driven tool for qualitative assessment. In this approach, results are presented as puzzles on the device graph, challenging users to infer the underlying structure of qubit pairs as noise progressively increases. The signature metric of QA is the threshold circuit depth at which noise overwhelms the ability of two-body observables to detect correlations, commonly referred to as the “Peak of Doom.” Experiments on IBM Quantum hardware demonstrate a substantial improvement in this threshold, increasing from approximately four on 16-qubit devices in 2018 to approaching 100 on 156-qubit devices in 2024.

Building on these foundations, this work introduces Mirror QA (MQA), a hybrid framework that integrates the creation of sparse entangled pairs into MRB circuits to enable localized analysis. This paper is structured as follows.

II. BACKGROUND

A. Mirror Randomized Benchmarking

Errors in quantum gates arising from decoherence, imperfect control, and crosstalk accumulate rapidly in multi-qubit circuits, necessitating scalable benchmarking protocols that quantify average performance without exponential resource overheads [4, 12]. Traditional Randomized scoring (RB) methods, such as Standard RB (SRB) [13, 14] and Direct RB (DRB), aggregate diverse error sources into a single infidelity metric by averag-

ing the ensemble over random Clifford circuits. These techniques excel in precision for small systems but falter at scale: SRB demands circuit depths scaling as $\mathcal{O}(n^2/\log n)$ for n qubits due to Clifford compilation [15–17], limiting implementations to $n \leq 3$ [18], while DRB, though streamlined, remains confined to $n \leq 5$ [19]. Meanwhile, volume-oriented benchmarks like quantum volume [20] and cross-entropy benchmarking [21] require classically intractable simulations beyond $n \sim 50$ [20–22]. This scalability chasm obscures holistic insights into many-qubit errors, particularly spatially correlated phenomena like crosstalk, which evade single- and two-qubit calibrations.

Mirror Randomized Benchmarking (MRB) was introduced as a protocol that redefines scalable quantum performance assessment, combining conceptual simplicity with broad applicability. MRB employs randomized mirror circuits to enable robust characterization of large-scale quantum processors. These are a customizable ensemble of Pauli-twirled, self-inverting layers—to estimate the average entanglement infidelity $\epsilon_\Omega = \mathbb{E}_\Omega[\epsilon(L)]$ of Pauli-dressed n -qubit Clifford layers under a user-specified distribution Ω over layer set $\mathcal{L}$. These circuits (Fig. ??) interleave $d/2$ pairs of layers L_i and L_i^{-1} (pink boxes, sampled from Ω) with $d+1$ uniform Pauli layers P_i (green) for twirling and initial/final single-qubit Cliffords F_0, F_0^{-1} (blue) for local randomization. By construction, ideal circuits yield a single, classically computable bit-string s_C , while errors manifest themselves as deviations in the output distributions. Execution involves sampling K circuits per even depth $d \geq 0$, running each N times, and computing the effective polarization

$$S = \frac{4^n}{4^n - 1} \left[\sum_{k=0}^n \binom{n}{k} \left(-\frac{1}{2}\right)^k h_k \right] - \frac{1}{4^n - 1}, \quad (1)$$

where h_k is the probability of Hamming distance k from s_C . Fitting $S_d \approx Ap^d$ yields $r_\Omega = (4^n - 1)(1 - p)/4^n \approx \epsilon_\Omega$, correcting for state preparation/measurement (SPAM) errors via binomial inversion. Under Markovian assumptions, theory proves exponential decay with $p^2 \approx \mathbb{E}_\Omega[\gamma_{L^{-1}}\gamma_L]$ (where $\gamma = 1 - 4^n\epsilon/(4^n - 1)$), with scrambling properties ensuring negligible error correlations. This yields reliable estimates even for non-uniform errors, with relative biases $|\delta_{\text{rel}}| \lesssim 0.2$ in physical regimes.

What elevates MRB is its good scalability and diagnostic power. Simulations on 225-qubit lattices (15×15 grid) with realistic Pauli errors (0.1 % single-qubit, 1 % two-qubit infidelities) confirm $r_\Omega \approx \epsilon_\Omega$ across 900 trials, unmasking quadratic crosstalk scaling absent in gate-local models (Fig.). On IBM Quantum devices (Quito and Rueschlikon), MRB benchmarks $n \leq 16$ qubits, matching DRB for $n \leq 5$ while exposing n -dependent divergences from calibrations quantifying crosstalk contributions invisible to isolated RB[9].

B. Quantum Awesomeness

Quantum Awesomeness (QA) [10] is a complementary benchmarking framework based on random circuits, proposed initially to visualize and interpret the behavior of quantum devices through human-interpretable correlation patterns. In its original formulation, QA applies random entangling layers followed by reconstruction attempts based on the observed output statistics. The protocol evaluates how effectively these correlations persist as the circuit depth increases, thereby providing an intuitive measure of the buildup and noise accumulation.

In contrast to MRB, QA focuses on the structure of correlations between qubit pairs. Metrics such as pairwise probability agreement (i.e., $p_j = p_k$ for entangled qubits) were used to infer entanglement fidelity and track the spread of randomness through the register. The degradation of these structured correlations with increasing circuit depth marks the transition from coherent quantum dynamics to noise-dominated behavior.

A notable feature of the QA framework is its gamified interpretation. Measurement statistics from each qubit are mapped to numbers or colors, and the challenge is to identify which qubits formed entangled pairs in the most recent circuit layer. At shallow circuit depths, this puzzle is straightforward, since correlations are strong and localized. As circuit depth increases, the spread of entanglement, cross-talk, and decoherence gradually obscures the correlation structure, making the puzzle progressively more difficult. The point at which meaningful structure disappears corresponds operationally to the onset of chaotic circuit behavior and the breakdown of reversibility. This framing provides an intuitive visualization of the loss of quantum coherence and the buildup of noise.

A later version of Quantum Awesomeness was introduced for utility-scale quantum devices [11]. In this updated approach, the increase in device size and circuit depth made the original probability-based pairing metric less effective: as noise grew with depth, many qubit probabilities clustered around 0.5, obscuring the underlying entanglement structure. To overcome this limitation, the updated QA framework replaced probability comparisons with *mutual information* as the primary metric of correlation. Mutual information captures both linear and nonlinear statistical dependencies between qubits, and thus remains sensitive even when marginal distributions become uniform. The updated game challenges players (or algorithms) to infer correlated qubit pairs from the mutual information matrix rather than from simple probability matching. This modification significantly enhances the interpretability of deeper circuits, mirrors the more complex entanglement structures achievable on modern hardware, and aligns the game with contemporary information-theoretic benchmarking methods.

C. Motivation for a Hybrid MQA-MRB Framework

MRB and QA emphasize different aspects of quantum processor behavior. MRB provides scalable, quantitatively rigorous fidelity estimates based on circuit reversibility, whereas QA offers a qualitative, human-interpretable view of the correlation structure and its degradation.

These complementary strengths motivate a hybrid framework that combines the structured reversibility of MRB with the additional quantitative and qualitative insights of QA. Embedding QA-style entangling layers around MRB mirror circuits creates a unified methodology that can probe entanglement dynamics, noise propagation, and reversibility within a single experiment.

The resulting Mirror Quantum Awesomeness (MQA) protocol synthesizes MRB with the updated QA methodology. The hybrid circuit structure incorporates QA-inspired entangling layers, MRB mirror blocks, and mutual-information-based analysis of correlation dynamics. The following section details this integration, describing the circuit construction, benchmarking parameters, and correlation metrics used throughout the study.

III. MIRROR QUANTUM AWESOMENESS

A. Structure Of The Circuit

In the randomized mirror benchmarking (MRB) framework, circuits are constructed from alternating layers of random single-qubit Pauli gates and randomly sampled Clifford-type circuit layers. The Pauli layers act as a scrambling mechanism as they introduce strong local randomization that prevents coherent errors from accumulating constructively over repeated circuit depth. As a result, coherent error contributions are effectively converted into stochastic-like behavior, leading to more stable decay curves and simplifying the interpretation of benchmarking data.

A width- w randomized mirror circuit of benchmark depth d typically consists of the following components:

1. An initialization layer preparing all qubits in the computational basis state $|0\rangle^{\otimes w}$.
2. A layer of uniformly random single-qubit Clifford gates, applied independently to each qubit.
3. A sequence of layers sampled from a specified distribution Ω , where each repeated unit consists of:
 - (a) a uniformly random single-qubit Pauli layer, followed by
 - (b) a circuit layer sampled from Ω , which may contain both single-qubit Clifford gates and multi-qubit entangling operations.

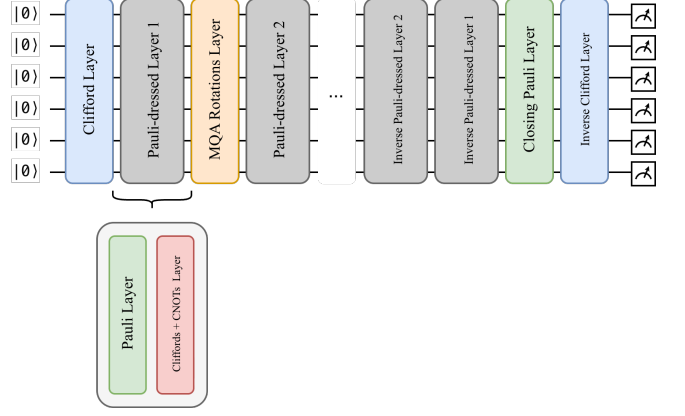


FIG. 1. MQA circuit structure: The yellow block represents the additional structured rotation layer inserted into the forward portion of the standard MRB circuit. All other layers correspond to the MRB architecture, consisting of an initial Clifford layer, a sequence of Pauli-dressed layers, their inverse counterparts, and a final closing Pauli layer

4. A final random Pauli layer applied before inversion.
5. A mirrored inverse sequence, where the sampled circuit layers are applied in reverse order and replaced by their inverses, while the Pauli layers are resampled independently.
6. A final inverse Clifford layer, which inverts the initial Clifford layer.
7. Measurement in the computational basis.

This construction ensures that the forward randomized circuit evolution is followed by a carefully constructed mirror portion, such that in the absence of noise the overall circuit ideally returns the system to its initial state.

Although MRB provides a scalable benchmarking backbone, it does not explicitly enforce tunable correlations or controlled entanglement structure across different circuit depths. To address this limitation, an additional structured rotation layer is incorporated into the MRB architecture, resulting in the modified circuit family referred to as MQA.

As illustrated in Fig. 1, the MQA circuit preserves the full MRB mirror structure, consisting of an initial single-qubit Clifford layer, a sequence of Pauli-dressed circuit layers, their corresponding inverse layers, and a final closing Pauli layer. The key modification is the insertion of an intermediate rotation layer within the forward half of the circuit, specifically between the first and second Pauli-dressed layers.

The inserted MQA rotation layer consists of deterministic single-qubit $R_x(\theta)$ rotations applied on the control qubits of all qubit pairs involved in the first-layer controlled-NOT (CX) operations. By varying the rotation angle θ , the degree of correlation introduced into the circuit can be systematically controlled. In particular, choosing $\theta = \pi/2$ generates maximally entangled pairs,

establishing strong correlations at the outset, whereas $\theta = 0$ introduces no additional entanglement and reduces the circuit back to the original MRB structure.

Intermediate values of θ , such as $\theta = \pi/4$, generate partially entangled states and enable exploration of regimes beyond purely Clifford dynamics. This is significant since it extends the benchmarking family to include controlled non-Clifford perturbations, allowing a more comprehensive characterization of error propagation and correlated noise effects under general quantum evolution.

B. Circuit Parameters and Modified Design

The MQA framework extends the mirror-circuit benchmarking protocol by refining several core parameters that control the circuit structure, data sampling, and analysis. The following subsections describe the primary parameters and methodological differences introduced relative to MRB.

a. Entangling Angle. The entangling angle defines the rotation applied in two-qubit gates. For Clifford circuits, primarily 0 and $\pi/2$ are used. The zero rotation generates no entanglement, while $\pi/2$ produces a maximally entangled pair.

In MQA, these angles were applied explicitly in the first and last layers of the circuits to control entanglement and facilitate simulations. It specifically inserts RX gates into the control qubits of any qubit involved in a CX in the first layer, leading to entanglement between all pairs in that layer.

b. Two-Qubit Gate Density. A key feature of randomized mirror benchmarking (MRB) circuits is the density of two-qubit entangling gates, since these gates typically dominate noise in near-term quantum devices. For a circuit C of width w and benchmark depth d , let α denote the total number of two-qubit gates. The two-qubit gate density is defined as

$$\xi = \frac{2\alpha}{wd}. \quad (2)$$

This can be interpreted by viewing the circuit as a $w \times d$ grid, where each two-qubit gate occupies two qubit-locations. For example, a density value of 0.25 implies that approximately one-quarter of all possible pairs of two-qubits are selected to form entanglement gates in each layer, which in practice results in about half of the qubits being involved in entanglement at a given time, providing a balanced trade-off between circuit depth and entanglement complexity.

In this work, the default density 0.25, as described in the MRB [9], is used in most of the analysis, but is later modified to study the effect of circuit topology. Controlling ρ allows one to tune circuit complexity and entanglement while maintaining high disorder, which is crucial for benchmarking quantum devices.

c. Sampling Algorithm. Randomized mirror circuits are generated by sampling layers from a defined set

L of n -qubit operations. Here, L consists of all parallel applications of CNOTs between connected qubits and all 24 single-qubit Clifford gates. The choice of sampling algorithm determines the structure and density of two-qubit gates within each layer, directly influencing circuit depth, entanglement structure, and error sensitivity. In this work, two types of layer samplers were employed: (i) the *Edge-Grab Sampler*, and (ii) the *Matching Sampler*.

1. Edge-Grab Sampler. The placement of two-qubit gates in randomized mirror circuits is determined using an *edge-grab sampler* parameterized by a target expected two-qubit gate density $\bar{\xi}$. The sampler selects a set of disjoint edges from the device connectivity graph, ensuring that no qubit participates in more than one two-qubit gate within the same layer.

The sampling procedure is defined as follows:

- (a) Initialize a candidate edge set $E = \emptyset$, and let E_r be the set of all edges in the hardware connectivity graph.
- (b) While E_r is not empty:
 - i. Select an edge v uniformly at random from E_r .
 - ii. Add v to E , and remove from E_r all edges that share a qubit with v .
- (c) Each edge in E is then included in the final sampled edge set independently with probability

$$p = \frac{w\bar{\xi}}{|E|}, \quad (3)$$

where w is the circuit width and $|E|$ is the size of the candidate set.

This method produces layers with a controllable expected number of two-qubit gates, allowing the overall circuit ensemble to achieve an average two-qubit gate density close to $\bar{\xi}$. The edge-grab sampler therefore provides a scalable and hardware-compatible approach for generating randomized circuit layers with tunable entangling complexity.

2. Matching Sampler. The matching-based sampling algorithm is constructed using a maximum-weight matching approach and was primarily used for analyzing device topology. Given a set of w qubits, their connectivity graph, and a desired two-qubit gate density ξ_s , this sampler proceeds as follows:

- (a) Perform a maximum-weight matching E on the connectivity graph, assigning random weights to edges to introduce stochasticity while ensuring that E consists of disjoint connected qubit pairs.

- (b) From the matched edge set E , select each edge with probability

$$p = \frac{w\xi_s}{2|E|}, \quad (4)$$

where $|E|$ is the number of matched edges. The selected edges represent the two-qubit operations for the given layer.

- (c) Assign random single-qubit Clifford gates to all remaining qubits not participating in two-qubit operations.

This generates layers with an expected two-qubit gate density of $\xi = \xi_s/2$. When these layers are interleaved with single-qubit Pauli layers, the overall expected density converges to ξ_s for large circuit sizes. In this work, the matching sampler was specifically used for topology analysis, focusing on loop and square configurations. The algorithm was evaluated using the performance metric, both with and without error-mitigation techniques, to assess the roles of connectivity and layout-dependent error behavior.

d. Parity-Based Measurement Analysis. In the benchmarking analysis, the rewrite-string method converts raw measurement bitstrings into a processed form that encodes the expected parity structure of the circuit after mirror operations. This processing allows bitstrings from mirrored subcircuits to be meaningfully compared across different configurations of qubit pairs.

In the modified implementation, explicit parity encoding is introduced:

- For every single qubit, the bit value is copied directly.
- For each two-qubit pair (q_0, q_1) , the bit corresponding to the lower-indexed qubit encodes the parity where the higher-indexed qubit in the pair is set to zero as a placeholder.

This ensures that the rewritten string represents pairwise parity information rather than raw qubit states, capturing correlations induced by entangling operations, particularly in the first and last layers of the circuit.

As a result, the processed bitstrings provide not only a reduction of artificial noise from redundant bits but also more consistent and physically meaningful benchmarking metrics, such as polarization or fidelity decay versus circuit depth, with better fits.

C. Mutual Information as an Extra Metric

Previous metrics used for QA (fuzziness, MWPM, and angle differences) [10] quantify the impact of noise across all qubit pairs when each pair is prepared with a different entangling angle. In the present MQA setting, however,

all entangled pairs share the same angle, so these metrics are not directly applicable. Mutual information is therefore introduced as a more suitable alternative. It provides a quantitative way to characterise different types of qubit pairs in this scenario. Since not every pair will be equally or maximally entangled, different pairs are expected to follow distinct trends. To interpret these trends, the formal definition of mutual information is first introduced, followed by the explanation of the different types of qubit pairs it helps to characterise.

Mutual information (I) [23] quantifies the amount of information shared between two variables X and Y . It is computed by subtracting the joint entropy of the two variables from the sum of their independent Shannon entropies

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (5)$$

Analyzing how mutual information evolves for each pair of qubits as the circuit depth increases can reveal how noise impacts the system and help identify built-in correlations. For a given pair of qubits, the marginal and joint distributions p_j , p_k , and $p_{j,k}$, respectively, are used to compute the corresponding Shannon entropies

$$I(j; k) = H(p_j) + H(p_k) - H(p_{j,k}). \quad (6)$$

The Shannon entropies of the measured outcome distributions are sufficient for the qubit pair characterization. Von Neumann entropies would require density matrices calculations involving measurements in multiple bases. If classical entropies already allow for the characterization, there is no need to account for multiple bases.

To better understand how these correlations arise and evolve, qubit pairs are classified into three distinct categories:

1. **Paired Edge:** Qubit pairs selected to be entangled by the newly inserted MQA layer.
2. **Spectator Edge:** Qubit pairs in which at least one qubit is connected to a paired qubit but is not itself part of an entangling operation.
3. **Isolated Edge:** Qubit pairs whose qubits are neither involved in nor adjacent to any paired edge.

To characterize the global behavior of each edge category, the mutual information is averaged over all qubit pairs within that category. For this purpose, the Mean Mutual Information (MMI) is used, which averages over pairs to yield a holistic value for each of the three categories. The MMI is close to 1 at the beginning for paired edges, since these correspond to the maximally entangled qubits, while the spectator and isolated edges have an approximated MMI value of 0.

As depth increases, noise begins to disrupt the correlations of paired edges and can also generate correlations in

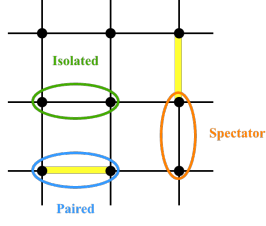


FIG. 2. **Types of Edges**

The yellow lines indicate qubit pairs that become entangled by the newly inserted MQA layer. Blue edges represent paired qubits, orange edges are the spectator qubits, and the remaining isolated edges are shown in green.

spectator and isolated edges. Consequently, the MMI for the paired edges starts to decay, while the MMI for the spectator and isolated edges grows, reaches a peak, and then drops again. Accordingly, noise first acts as a correlation builder, but eventually becomes strong enough to wash out all correlations, causing the MMI to decay.

Figure 3 shows the behaviour of the different types of edges under different rotation angles and different entangling densities on a customised noisy backend simulator for a better understanding of this decay. A custom noisy backend has been chosen over a stabilizer backend to illustrate the build-up of correlations under different entangling angles, since the stabilizer backend is based on Clifford operations and can only support multiples of 0 and $\pi/2$ as input angles. Eight qubits have been used for simplicity, as this customised noisy backend significantly increases runtime with increasing qubit count. If the initial entangling angle is $\theta = \pi/2$, the paired edges become maximally correlated. This leaves no room for correlations to build on neighbouring spectator edges, so the MMI of the spectator edges remains pinned at zero. For $\theta = \pi/4$, there is room for noise to build correlation on the spectator edges, which is why their MMI shows a peak followed by a decay. In contrast, the isolated edges always exhibit a peak regardless of the initial entangling angle, since they are never confined to entangled states, and noise can always build correlations on them. When $\theta = 0$, both the spectator and paired edges effectively behave like isolated edges, showing a peak and returning to the MRB-like behaviour where no additional rotation gates are applied. The two-qubit gate density ρ also influences how correlations develop. At maximal density, no isolated edges exist because the distribution of entangling gates is saturated. Reducing the density below this maximum reintroduces the isolated-edge curve.

Figure 4 shows the stabilizer simulator running on the full IBM Fez coupling map, comprising 156 qubits. Since only Clifford gates are used in this backend, the entangling angle is set to $\pi/2$. The strong correlations on the paired edges leave no room for correlation build-up on the spectator edges, mirroring the behavior observed in the custom noisy backend. As expected, the isolated edges still exhibit a peak.

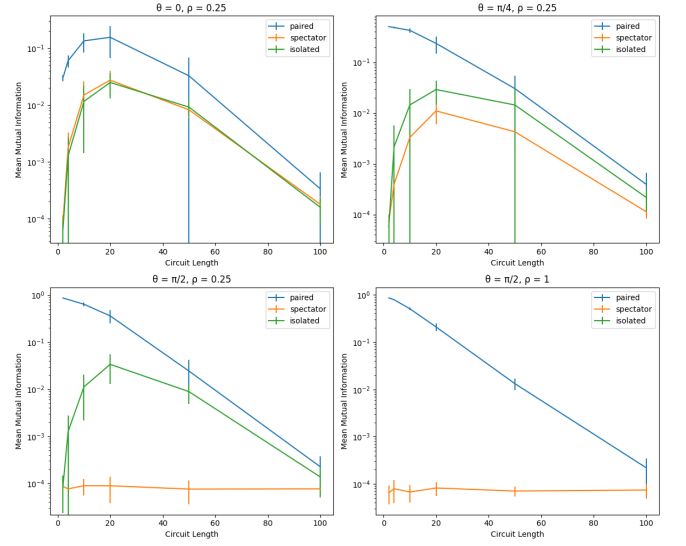


FIG. 3. **MMI – Custom Noisy Backend**

Dependence of the Mean Mutual Information (MMI) on the initial entangling angle θ and entanglement density ρ for each edge type in an 8-qubit system, with `samples = 20` and `shots = 10000`.

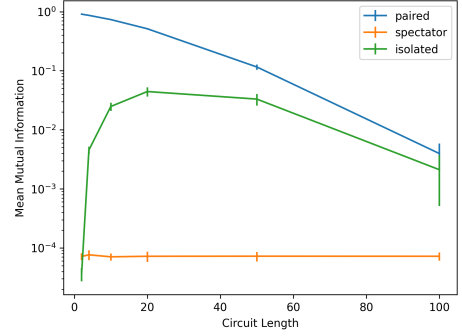


FIG. 4. **MMI – Stabilizer Backend**

MQA simulations of 156 qubits on the IBM Fez topology with `samples = 20` and `shots = 10000`. The initial entangling angle is fixed at $\theta = \pi/2$, consistent with the stabilizer backend that allows only Clifford gates.

The depth at which the MMI peaks depends on the depolarizing noise. As the noise increases, the peak shifts to the left, since stronger noise accelerates both the build-up and the decay of the MMI curve. The peak position does not depend on the qubit count, since noise propagation is predominantly local.

IV. RESULTS

A. Verification

To assess the impact of incorporating additional entangling layers into the Mirror Randomized Benchmarking (MRB) framework, the protocol is evaluated using both

numerical simulations and real quantum hardware. This augmentation introduces *tunable entangling slices*, parameterized by an initial angle θ , into the mirror-circuit structure. Changing θ , the modified MRB systematically probes the buildup of entanglement while preserving the exponential decay analysis of the core that underpins the standard MRB.

Performance is quantified using the *Entanglement Infidelity* (EI), which serves as the central benchmarking metric. The EI values are compared between multiple MQA simulations and hardware executions. In numerical simulations, both MRB and MQA were evaluated using the IBM Fez reference layout with qubit counts ranging from $n = 2$ to $n = 156$. The simulations were executed using a stabilizer-based simulator, enabling controlled benchmarking runs across large system sizes. Realistic noise models were incorporated, including variable single-qubit depolarizing noise and two-qubit depolarizing noise, to emulate experimentally relevant error sources. When standard MRB analysis was applied at higher qubit counts, the resulting decay-curve fitting exhibited poor convergence and instability which in this work was mitigated through the introduction of a parity-based measurement analysis method, described in Section 3. This method is conceptually motivated by the rotation layer used in the MQA construction, as it evaluates the parity of bit-string outcomes rather than performing direct bit-by-bit comparisons. The parity-resolved approach significantly improves the quality of decay fitting at large system sizes, yielding stable EI extraction across the full simulated range. The figure below shows the simulation plots illustrating the contrast between the degraded fitting obtained with the conventional method and the improved behavior enabled by the parity-based analysis.////

Across all simulated system sizes and circuit depths, the mean absolute difference in the extracted Entanglement Infidelity (EI) between MRB and the modified MQA protocol was // which corresponds to approximately 8.49 //— of the global average Entanglement Infidelity value of 0.02189. This small deviation across the full qubit range indicates that the MQA augmentation introduces only a negligible perturbation to standard MRB estimates in simulation. Consequently, MQA preserves the scalability and theoretical fidelity of MRB.

On real quantum hardware, MQA experiments were performed for $\theta = 0$ and $\theta = \pi/2$ on the IBM Fez and IBM Kingston processors and compared directly with standard MRB. The extracted Entanglement Infidelity (EI) values show close agreement across both devices. For IBM Fez, the average EI difference between MQA and MRB is approximately 0.6%, while for IBM Kingston the corresponding difference is approximately 4.7%.

In both cases, the observed deviations between MQA and MRB remain small relative to the absolute EI values and lie within the same order of magnitude. These findings indicate that MQA provides EPC estimates broadly consistent with MRB, with only mod-

est protocol-dependent variation. Such variation is expected on experimental hardware, where EPC extraction depends on the device's instantaneous calibration state, temporal noise fluctuations, and the number and spatial distribution of simultaneously active qubits.

During the Kingston experiments in particular, several qubits near the processor center remained inactive, producing altered effective connectivity and a nonuniform noise landscape. This operating condition motivated a closer comparison between EI estimates obtained from MQA and MRB under non-ideal hardware configurations. Despite these factors, the differences between the two protocols remain small, demonstrating that MQA and MRB yield effectively equivalent characterizations of the average entanglement Infidelity. Therefore, the quantitative EI differences observed between Kingston and Fez are attributed primarily to time-dependent hardware characteristics rather than systematic discrepancies between the benchmarking methodologies.

The following figures show the benchmarking results obtained on the same day for a 20-qubit configuration at $\theta = \pi/2$, illustrating MRB, MQA on IBM Fez and IBM Kingston, respectively.

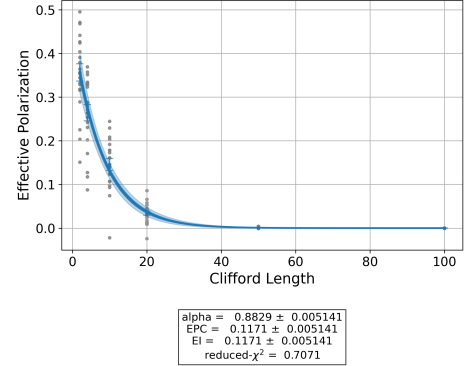


FIG. 5. MRB - IBM Fez
20 qubits

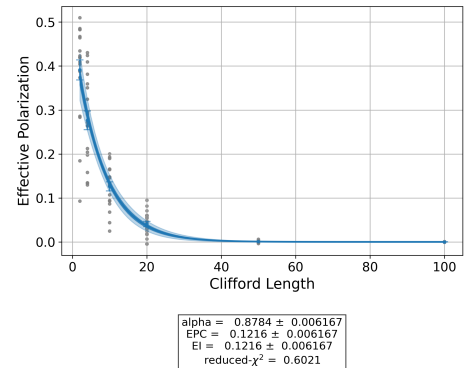


FIG. 6. MQA - IBM Fez, $\theta = \pi/2$
20 qubits

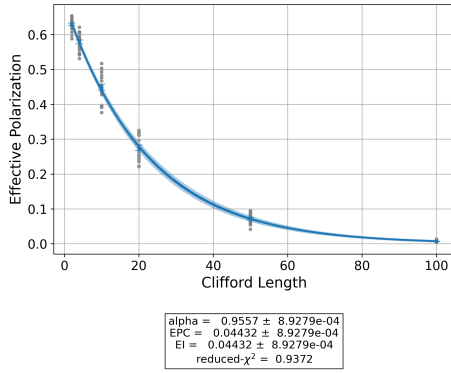


FIG. 7. MQA - IBM Kingston, $\theta = \pi/2$
20 qubits

B. EI

C. Peak of Doom

Figure 8 shows how MQA performs on IBM Fez real hardware for the entire device when the initial entangling angle is set to $\pi/2$. Similarly to what was observed for the customised noisy backend and the stabilizer backend, the spectator edges are unable to build correlations, and only the isolated edges show a peak. The real hardware therefore follows the expected behaviour. The peak lies between lengths 20 and 50, and the mutual information of the paired qubits decays to 0.1 at around length 50. Figure 9 shows MQA running again on the full IBM Fez device, with the initial entangling angle set to $\pi/4$. This causes the spectator edges to experience correlation build-up as expected, once again matching the behaviour observed with the customised noisy backend.

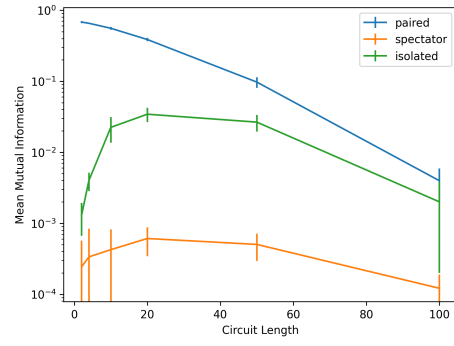


FIG. 8. MMI - IBM Fez for $\theta = \pi/2$
MQA experiments of 156 qubits on IBM Fez with **samples** = 20 and **shots** = 10000. The initial entangling angle is fixed to $\theta = \pi/2$.

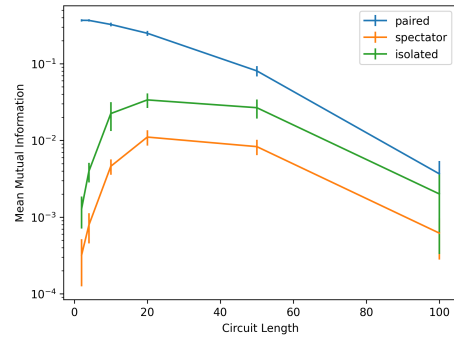


FIG. 9. MMI - IBM Fez for $\theta = \pi/4$
MQA experiments of 156 qubits on IBM Fez with **samples** = 20 and **shots** = 10000. The initial entangling angle is fixed to $\theta = \pi/4$.

Appendix A: Terms

RB (Randomized Benchmarking).
SRB (Standard Randomized Benchmarking).
DRB (Direct Randomized Benchmarking).
MRB (Mirror Randomized Benchmarking).
QA (Quantum Awesomeness).
MQA (Mirror Quantum Awesomeness).
EPC (Error Per Clifford).

I (Mutual Information). Quantifies the amount of information shared between two variables and defined as $I(X;Y) = H(X) + H(Y) - H(X,Y)$. Used here to track pairwise correlations versus circuit depth.

MMI (Mean Mutual Information). The mutual information averaged over a specified edge category (paired/spectator/isolated).

Appendix B: Extra Figures

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